

VibeQueue

Lab 1 Outline

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1. Introduction

In the modern landscape of the hospitality business there is more to consider than just the quality of the menu and service, with the atmosphere of the location and the “vibe” of the setting being crucial to ensure customer satisfaction. These factors will influence potential customers to enter the location as well as determine the time that is spent in the venue and customers who stay longer tend to purchase more things. One area that is lacking for smaller scale venues is a cost effective way to control the musical aspect of their atmosphere. There are over a million independent locations selling food and drinks that lack a customer interactable method to control music within affordable ranges. These businesses are in need of such a service as that

The currently established methods that can be used to control the music of a location are divided between the expensive proprietary methods of industry leaders like TouchTunes and RockBot or the slow manual workarounds that can be done by the staff. TouchTunes utilizes expensive jukebox hardware with an upfront installation cost that is beyond the acceptable range for most small venues. Some small locations use personal musical streaming accounts and open themselves to legal liabilities due to terms of service preventing private commercial use.

These conditions mean that there is a market gap in this area and this product will be able to fill this gap without much issue. VibeQueue will be low cost and fit within the limitations of a small budget local business. Additionally, it will be hardware agnostic and able to function without any problems on any customer device with internet connection. The service will be customer-facing and maintain legal compliance. It will need to be possible for the owner and operator to veto a song selection and constraints to restrict song genres will need to be possible as well.

It is our privilege to introduce VibeQueue to solve these problems and provide customers with a way to manage their venue’s musical accompaniment to assist in maintaining the atmosphere of the location. This product will be easy to utilize and cost effective for low budget small businesses. It is a seamless and legally permissible method that can be used to offer an interactive experience at places like your local coffee shop or other small businesses. The guiding statement of this service is “Your Vibes, Managed” and we look forward to aiding small businesses in that management.

2. VibeQueue Product Description

VibeQueue is a digital music service that is intended for small, independent venues and provides a cost-effective method to manage atmosphere without traditional jukeboxes or expensive proprietary systems. This will allow businesses to let customers influence the background music from their phone and bridge the gap between customers and the business. This will occur while the product ensures the “vibe” stays consistent.

3. Identification of Case Study

The main case study for VibeQueue is a single-location independent coffee location called Brewpoint Café which faces two issues that would be related to this product. The productivity of staff is hindered by customers requesting song changes and the owner has difficulty maintaining the atmosphere while accommodating these requests.

This product is being developed for businesses like this where there is a need to automate these functions, but established industry names are prohibitively expensive. By testing at this location, it will prove that the business can maintain the intended “vibe” while increasing customer involvement and generating small amounts of revenue at the same time.

4. Prototype

The prototype will differ from the product in several ways. The prototype may use a simulated transaction system that differs from the product. The prototype is likely to be less advanced in terms of scalability with a focus on a single location. The analytics of the prototype will be minimal while the product will have a significantly more advanced suite of analytics information. Moderation of the prototype will be significantly less extensive than the completed product.

5. Glossary

- **API (Application Programming Interface):** a set of rules and protocols that allows different software applications to communicate with each other
- **ASCAP (American Society of Composers, Authors, and Publishers):** a U.S. performing rights organization that licenses music for commercial use
- **BMI (Broadcast Music Inc.):** a U.S. performing rights organization that manages commercial music licensing on behalf of songwriters and publishers
- **Business Control Center:** VibeQueue’s owner facing web dashboard for configuring vibe constraints, managing the song queue, and viewing analytics
- **Hardware Agnostic:** software designed to run on any existing device without requiring specific or proprietary hardware

- **LAMP:** an open source web development stack comprised of Linux (OS), Apache (web server), MySQL (database), and PHP (scripting language)
- **MFCD (Major Functional Component Diagram):** a visual representation of a system's major components and the data flows between them, as defined in CS 410
- **QR Code (Quick Response Code):** a type of two dimensional matrix barcode that can be scanned by a smartphone camera to open a URL or application
- **SESAC (Society of European Stage Authors and Composers):** a U.S. performing rights organization providing commercial music licensing
- **SoundMachine:** a third party commercial music streaming service that holds ASCAP/BMI/SESAC licenses; used by VibeQueue for legal audio playback in venues
- **Soundcharts API:** a third party music analytics API that provides commercially safe song metadata including genre, tempo, mood, popularity, and explicit content flags
- **Vibe Engine:** VibeQueue's proprietary, in house song selection and validation algorithm; evaluates customer song requests against venue's configured constraints before allowing them into the queue

6. References

Rockbot. (n.d.). *Rockbot business music platform*. <https://rockbot.com/go>

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SoundMachine. (n.d.). *Commercial music streaming for businesses*. <https://sound-machine.com>

TouchTunes. (n.d.). *Our story*. <https://www.touhtunes.com/about/our-story>